

MATMEK4270 Assignment 1

SANDER OPPEN

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1.2.3. Exact solution

We begin by finding the partial derivatives for u , starting with t .

$$\begin{aligned}\frac{\partial u}{\partial t} &= e^{i(k_x x + k_y y)}(-i\omega)e^{-i\omega t} = -i\omega u \\ \frac{\partial^2 u}{\partial t^2} &= (-i\omega)^2 u = -\omega^2 u.\end{aligned}$$

Next, for the derivatives for x and y (I have skipped some steps when calculating, they are the same as the previous scenario).

$$\begin{aligned}\frac{\partial u}{\partial x} &= ik_x u \\ \frac{\partial^2 u}{\partial x^2} &= (ik_x)^2 u = -k_x^2 u. \\ \frac{\partial u}{\partial y} &= ik_y u \\ \frac{\partial^2 u}{\partial y^2} &= (ik_y)^2 u = -k_y^2 u.\end{aligned}$$

Putting all these into the wave equation, we get

$$-\omega^2 u = c^2(-k_x^2 u - k_y^2 u). \tag{1}$$

Since u has no zeroes, we know that the wave equation holds if and only if we have $\omega = \pm c\sqrt{k_x^2 + k_y^2}$.

1.2.4. Dispersion coefficient

We begin by finding the required values for u_{ij}^n , starting with $n + 1$ and $n - 1$.

$$\begin{aligned} u_{ij}^{n+1} &= e^{i(kh(i+j) - \tilde{\omega}(n+1)\Delta t)} = e^{-i\tilde{\omega}\Delta t} u_{ij}^n \\ u_{ij}^{n-1} &= e^{i\tilde{\omega}\Delta t} u_{ij}^n. \end{aligned}$$

Next, for $i \pm 1$ and $j \pm 1$ (I have again skipped some steps when calculating, they are the same as the previous scenario):

$$\begin{aligned} u_{i+1,j}^n &= e^{ikh} u_{ij}^n \\ u_{i-1,j}^n &= e^{-ikh} u_{ij}^n \\ u_{i,j+1}^n &= e^{ikh} u_{ij}^n \\ u_{i,j-1}^n &= e^{-ikh} u_{ij}^n. \end{aligned}$$

Putting all these into equation (1.3) from the problem statement, we get

$$\frac{e^{-i\tilde{\omega}\Delta t} - 2 + e^{i\tilde{\omega}\Delta t}}{\Delta t^2} = \frac{2c^2}{h^2} (e^{ikh} - 2 + e^{-ikh}). \quad (2)$$

Using $e^{ia} + e^{-ia} = 2\cos(a)$, we get

$$\frac{2\cos(\tilde{\omega}\Delta t) - 2}{\Delta t^2} = \frac{2c^2}{h^2} (2\cos(kh) - 2),$$

and by further using $\cos(a) - 1 = -2\sin^2(\frac{a}{2})$, we get

$$\begin{aligned} -\frac{4}{\Delta t^2} \sin^2\left(\frac{\tilde{\omega}\Delta t}{2}\right) &= -\frac{8c^2}{h^2} \sin^2\left(\frac{kh}{2}\right) \\ \sin^2\left(\frac{\tilde{\omega}\Delta t}{2}\right) &= \frac{8c^2\Delta t^2}{4h^2} \sin^2\left(\frac{kh}{2}\right) \\ \sin^2\left(\frac{\tilde{\omega}\Delta t}{2}\right) &= 2C^2 \sin^2(kh/2). \end{aligned}$$

In the last step we have used the substitution $C = \frac{c\Delta t}{h}$. For $C = \frac{1}{\sqrt{2}}$, this becomes

$$\sin^2\left(\frac{\tilde{\omega}\Delta t}{2}\right) = \sin^2\left(\frac{kh}{2}\right). \quad (3)$$

From this, we know that

$$\begin{aligned} \frac{\tilde{\omega}\Delta t}{2} &= \frac{kh}{2} \\ \tilde{\omega} &= \frac{kh}{\Delta t}. \end{aligned}$$

Going back to our constant C , we can express Δt as

$$\Delta t = \frac{h}{\sqrt{2}c}$$

Putting this into our previous expression for $\tilde{\omega}$, we get

$$\tilde{\omega} = \sqrt{2}ck.$$

Looking back at what we found in the previous problem, by setting $k_x = k_y = k$, we get that

$$\omega = c\sqrt{2k^2} = \sqrt{2}ck = \tilde{\omega}.$$